CHAPTER 21

Geosynchronous-satellite control-system design

21.1. Space story

Shortly after founding Princeton Satellite Systems, Doug Freesland contacted me and asked if we could design a momentum-bias control system for their new communications satellite, Indostar-1, for Indonesia. I said we could for 750$USD. He said they only had 230$USD. Since it was a cool project I said sure and we proceeded, with the help of engineers at his company, probably the lowest-cost communications-satellite control system on record.

21.2. Introduction

This chapter describes how to create a control-system design for a geosynchronous communications satellite. It describes how to interpret customer requirements, select actuators, and sensors, and organize the design. The design that is described flew on the Indostar-1 communications satellite.

21.3. Requirements

A simplified set of requirements for a communications satellite might be:

1. The beam center shall not deviate from nominal by more than 0.2 deg.
2. The spacecraft shall stay within 0.1 deg of the station for its entire life.
3. The spacecraft's life shall be greater than 10 years.
4. No single-point failure shall cause a pointing error of greater than 0.2 deg.
5. The payload consists of a global beam pointing at nadir. This is a beam that covers a very wide area (for example, the continental United States). In addition, there is a single-spot beam.

These are a minimal set of requirements and give the designer a lot of flexibility. However, there is usually a need to meet these requirements with the lowest possible cost.

The requirements only cover mission-orbit performance. It is also necessary to get the satellite to its mission orbit. If the satellite can ride on a Proton launch vehicle, it may not be necessary to worry about transfer orbit. Future spacecraft that use electric propulsion for transfer orbit will also not have a distinct transfer-orbit mode. Otherwise, there are three distinct modes:

1. Transfer orbit;

ADCS - Spacecraft Attitude Determination and Control
https://doi.org/10.1016/B978-0-32-399915-1.00033-4

2. Acquisition;
3. Mission orbit.

Transfer-orbit and acquisition requirements, besides getting the satellite safely to its mission orbit, are to minimize the fuel consumption since fuel is usually the life-limiting factor in a satellite design. Mission-orbit mode is often divided into station-keeping mode and normal mode.

21.4. The design process

The following design tasks can generally be conducted in parallel:

1. Simulation design;
2. Transfer-orbit control design;
3. Mission-orbit control-system design.

It is important to start the simulation design right away since simulations take a long time to test and debug. One generally should start by building the simplest possible rigid-body simulation with ideal sensors and actuators. The fidelity of the simulation can always be increased later as the design progresses.

21.5. Mission-orbit design

For this design, electric-heater-augmented hydrazine thrusters (electrothermal) will be used for inclination control, which is known as North/South station keeping. These thrusters are cost effective because inclination control requires a significant mass of propellant and the higher specific impulse of these thrusters (due to the electrical augmentation) reduces fuel consumption. These thrusters typically have thrust levels in the 0.5-N range. Attitude control thrusters will be needed to control the attitude during these maneuvers since the thrusters used for station keeping cannot be pulsed.

This leads to a decision on the mission-orbit architecture. There are three choices:

1. Spin-stabilized;
2. Momentum-bias;
3. Three-axis operation.

The first two are identical except that in the first the spinning part has larger inertia than the despun part. In the second, the opposite is true. The major advantage of a high-momentum design is that it provides passive attitude control. Although inertially fixed torques will precess the spin-axis, the momentum in the system will resist those torques. In addition, the momentum in the spacecraft makes it resistant to the consequences of thruster failures.

Another advantage is that since the momentum makes it resistant to disturbances, it may be possible to forgo a yaw sensor. It is easy to sense roll and pitch with an Earth sensor. Yaw is more difficult. It can be sensed intermittently with a Sun sensor, or continuously with a star sensor or Sun sensor/gyro combination. Forgoing a yaw sensor can save a lot of money and weight. For control, we can use thrusters, magnetic torquers, solar pressure, or wheel pivoting. For sensing, we will use an Earth sensor only during the mission orbit.

A three-axis design would use pivoted or fixed reaction wheels or thrusters for control. Reaction wheels require some mechanism for unloading momentum from the system. This can be done with thrusters, magnetic torquers, or solar pressure. Yaw sensing is required if there is no bias momentum since a body-fixed yaw disturbance cannot be sensed in roll and without yaw sensing, this would cause the attitude to diverge. On this spacecraft, the only yaw requirement is levied by the spot beam. A yaw error will translate into

$$\Delta Az = 0.035\Delta_{yaw} \tag{21.1}$$

$$\Delta El = 0.035\Delta_{yaw} \tag{21.2}$$

Thus we can tolerate much larger yaw errors than roll or pitch errors. This is because this mission does not have a direct yaw requirement, the beam is circular and the yaw error has a gain of 0.035 when applied to azimuth or elevation. This means that we can probably do without yaw sensing if we use a spinner or momentum-bias design. This might not be the case if we had crosslinks to other satellites or tighter spot beam-pointing requirements.

At this point we have a number of trades to make. Some of the possible configurations are shown in Table 21.1.

Table 21.1 Possible configurations.

Design	Roll/Yaw Control	Pitch Control	Roll/Yaw Unloading	Pitch Unloading
Spinner	Thrusters	Spin motor	N/A	Thrusters
Momentum Bias	Thrusters	MWA Motor	N/A	Thrusters
Momentum Bias	Magnetic Torquers	MWA Motor	N/A	Thrusters
Momentum Bias	Solar Pressure	MWA Motor	N/A	Solar Pressure
Momentum Bias	Pivots	MWA Motor	Magnetic Torquers	Thrusters
Momentum Bias	Pivots	MWA Motor	Thrusters	Thrusters
3-Axis	Reaction Wheels	Reaction Wheel	Thrusters	Thrusters
3-Axis	Reaction Wheels	Reaction Wheel	Magnetic Torquers	Thrusters

A careful cost trade-off should be made before selecting the configuration. For the sake of this example, assume that this has been done and the momentum bias with thruster roll/yaw control has been chosen. No yaw sensor will be used. The roll and pitch axes will be sensed using an Earth sensor. Earth sensors can be either static, consisting of a ring of thermopiles in the focal plane of the sensor, or scanning, which has an oscillating mirror that sweeps the image of the Earth across pyroelectric detectors or bolometers. The choice is one largely of cost.

A station-keeping control system will also be needed. Thrusters will be used as actuators but the choice of sensors is still open. Since the disturbances will be larger (tenths of μN versus hundreds of μN) a faster-acting control system will be needed. Earth sensors tend to be noisy, thus it is difficult to differentiate the signal to get a rate measurement for a high-bandwidth control system. As a consequence, it is a good idea to use gyros to produce rate information for station keeping. Since the gyros will only be used during station keeping, they need not be long-life gyros. While the Earth sensor can still be used for angle information in pitch and roll, the yaw gyro output must be integrated for yaw. This cannot give an absolute yaw measurement, only one relative to the original yaw estimate when the gyro was initialized.

21.6. The geometry

The orbital geometry of a spacecraft with its solar arrays deployed is shown in Fig. 21.1. The solar arrays are deployed from the North and South faces of the spacecraft (which is positive and negative orbit normal). When stowed, the array panels will lie flat on these surfaces. The East and West faces require space for thrusters to perform East/West station keeping. Consequently, the apogee kick motor (AKM) has to be on the anti-Earth face. The spin-axis will run through this face and point out the front of the nozzle. The Earth sensors will be on the nadir face. The antenna may get in the way of the sensors and cutouts in the dish (which should not pose much of an antenna problem if they are much smaller than the wavelength) may be necessary. During the AKM burn, the spin-axis will point at an angle from the Earth's equatorial plane roughly equal to the negative of the transfer-orbit inclination. After the AKM burn, it will be necessary to get this axis pointing at the Earth and rotating about the pitch-axis at orbit rate.

21.7. Control-system summary

The control system is summarized in Table 21.2.

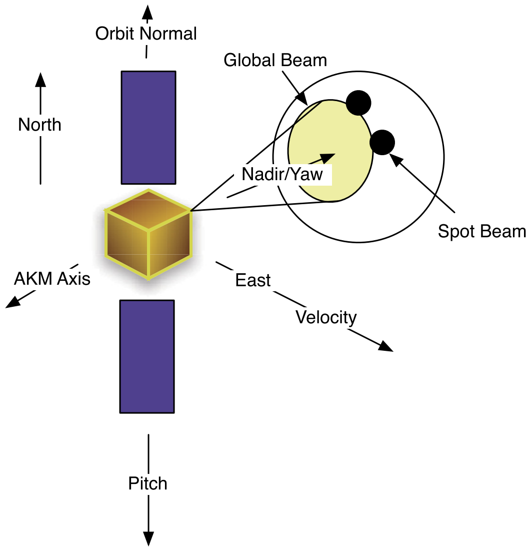


Figure 21.1 Orbital geometry.

Table 21.2 Control-system summary.

Mode	Actuator	Sensors
Transfer Orbit	Thrusters	Sun sensor and Horizon sensors
Acquisition	MWA motor	Earth sensor (for pitch)
Mission-Orbit Normal	MWA motor and Thrusters	Earth sensor
Mission-Orbit Station keeping	Thrusters	Gyros and Earth sensor

21.8. A mission architecture

The requirements: form a geosynchronous spacecraft with a mission life of 10 years. It will have a global beam that must point to nadir to within 0.2° and a one-spot beam. The goal is to design the least-expensive spacecraft that can achieve these requirements and make some money doing it! There are three phases to this mission:

1. Transfer orbit;
2. Acquisition;
3. Mission orbit.

During transfer orbit, the spacecraft is spinning about its major axis. This makes it passively stable. Reorientations are done using thrusters that precess the spin-axis. Transfer-orbit control is covered in Chapter 20.

During the mission orbit, the satellite is stabilized using a momentum wheel. The momentum in the wheel is sufficiently high so that the spacecraft is dual-spin stable. Control may be accomplished using either magnetic torquers and the momentum-wheel motor, or with thrusters. Station-keeping maneuvers are performed using thrusters. The transition between the spinning and momentum-bias states is done with the dual-spin turn. The transfer orbit, deployment, and mission orbit configurations are shown in Fig. 21.2.

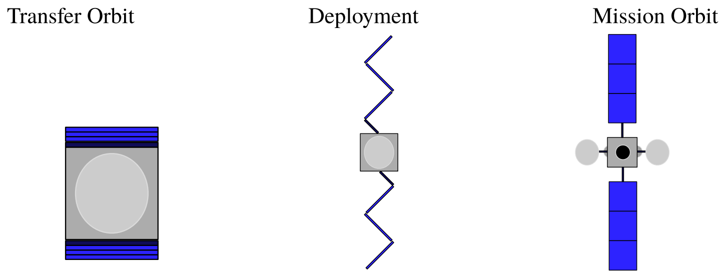


Figure 21.2 Spacecraft-configuration diagrams.

21.9. Design steps

Although a controls engineer will be concerned about such things as control-system stability margins, transient response, etc. the customer is only concerned about whether the satellite meets the requirements for beam pointing and spacecraft life. Spacecraft life is determined by the life of the components and by the amount of fuel the spacecraft carries. Component-life issues are addressed by carrying redundant components as necessary. The pointing budget and propellant budgets must be computed and updated regularly as the spacecraft design evolves.

21.10. Spacecraft overview

The spacecraft is illustrated in Fig. 21.3. Some of the features are listed below.

The spacecraft has radiators on the North and South faces. The solar wings have composite backings. This causes the wings to bend either towards or away from the Sun. In this case, the curve is approximated by a five-degree cant of the top two meters of the wings.

The geometric and surface properties are given in Table 21.3. The residual dipole is body fixed in an arbitrary direction.

The spacecraft layout was designed so that the spacecraft would be a major-axis spinner during transfer orbit and be nearly symmetrical about the Z-axis. In other words, the X-axis inertia and Y-axis inertia is nearly the same during transfer orbit.

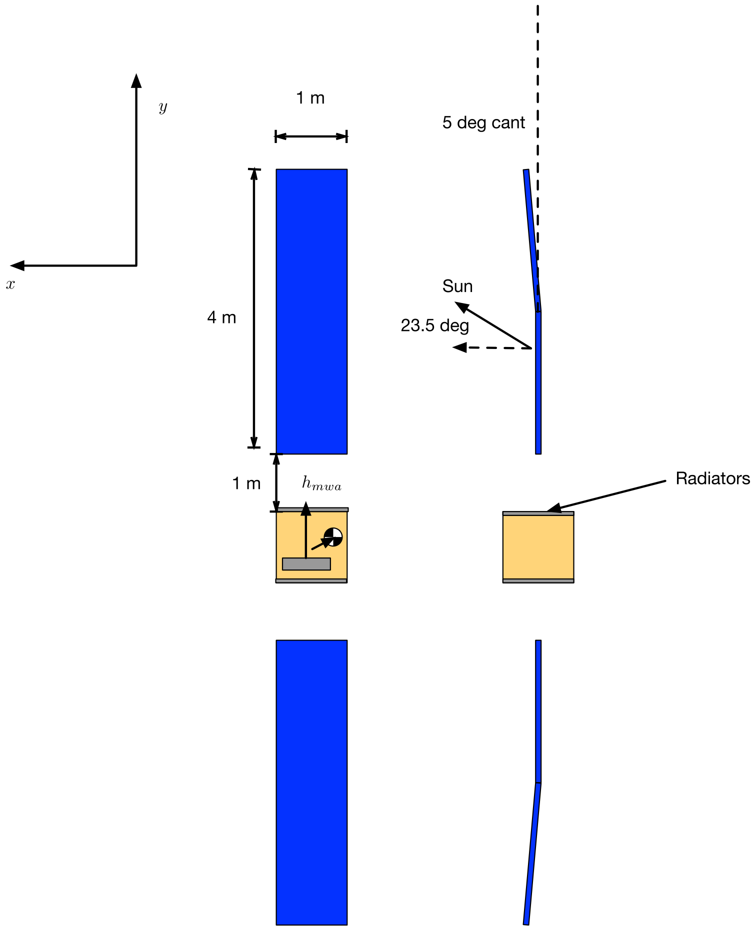


Figure 21.3 Close-up of spacecraft showing the Sun geometry.

This was accomplished by splitting the electronics into four boxes and locating them as illustrated above. Laying out a spacecraft so that it has good thermal properties, is easy to assemble, and is balanced is very difficult and expensive. A three-axis satellite stabilized in transfer orbit and did not spin at a high rate would not have to be so carefully balanced.

The antenna beams point at nadir, nearly $+Z$ in this picture. Their angle is dependent on the location of the feeds that are on the nadir panel. The fuel supply is split into two half-systems for redundancy. Two tanks are located on the East and West faces.

The thrusters are located at the corners of the box on the East, West, and North faces. The electrothermal hydrazine thrusters are clustered around the Y-axis on the

Table 21.3 Spacecraft properties.

Parameter	Value
Total Power	2077.8 W
Radiator Power	415.6 W
Solar-Array Area	8.0 m^2
Radiator Area	2.0 m^2
Residual Dipole	[2.9 2.9 2.9] ATM^2
Foil Optical [absorbed;specular;diffuse]	[0.00 0.29 0.71]
Radiator Optical [absorbed;specular;diffuse]	[0.00 0.69 0.31]
Cell Optical [absorbed;specular;diffuse]	[0.75 0.17 0.08]
Center-of-Mass	[0.03 0.02 -0.01] m
Radiator Torque	[0.01 -0.01 0.00] micro-N m

North face. Their exact location and cant angles are set to minimize losses due to plume drag. The plumbing is designed so that if the half-system fails the other half-system can perform all required maneuvers. A fixed-momentum wheel was chosen for simplicity. Pivoted wheels (either single or double) are more flexible but heavier and more expensive. They can provide higher control authority for nutation damping than the magnetic torquers.

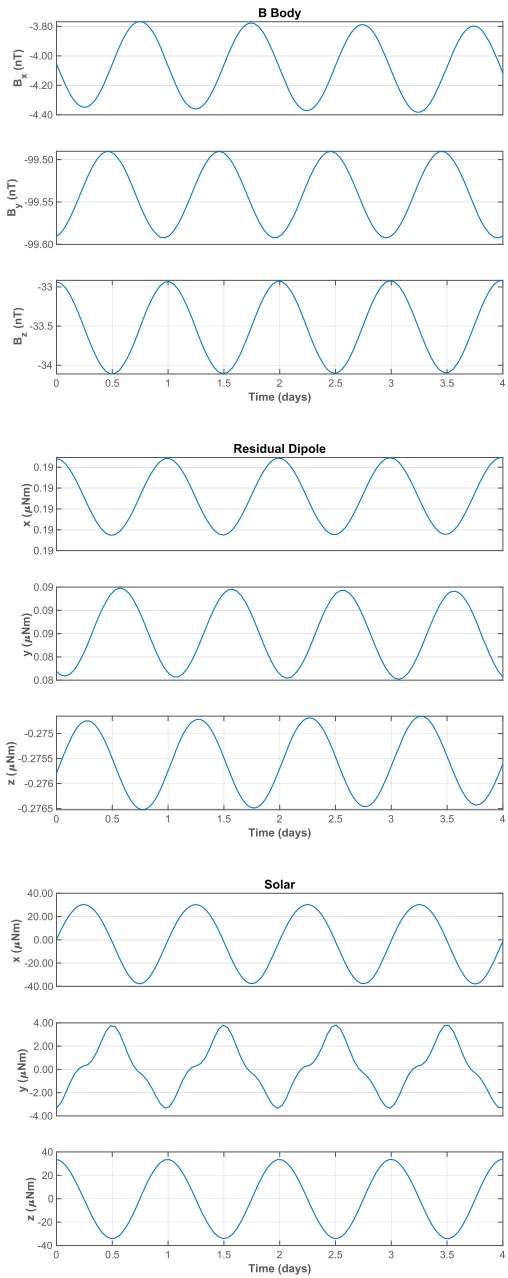
The propulsion system is a blowdown system. The tanks are filled with fuel and pressurized with helium. The mass and volume of helium determine the blowdown ratio, that is the ratio of initial pressure to empty pressure. For this spacecraft, we choose an initial pressure of 2 413 250 N/m^2 (350 psia) and a final pressure of 689 500 N/m^2 (100 psia).

The thruster layout is illustrated in Fig. 21.4. Thrusters 1 through 12 are monopropellant hydrazine thrusters with 5 N maximum thrust at a tank pressure of 2 413 250 N/m^2. Thrusters 12 through 16 are electrothermal hydrazine thrusters with a maximum thrust of 2 N at a tank pressure of 2 413 250 N/m^2.

21.11. Disturbances

The major disturbance sources on the spacecraft are solar pressure, residual dipole, and thermal torques. The model shown in Fig. 21.3 is used. Example 14.2 computes the disturbances for four orbits during the summer solstice. The Sun is 23.5 degrees out of the orbit plane. This causes a disturbance due to the canted solar wing panels. The solar panels always face the Sun, producing a xz harmonic disturbance at orbit rate. The rotation of the Sun around the bus causes the higher harmonics in y.

```matlab
%% Setup
SetupDisturbances;

%% Disturbance model
t       = linspace(0,4*24*3600,500);
[r,v]   = RVOrbGen([42167 0 0 0 0 0],t);
jD      = jD0 + t/86400;

bECI    = BDipole(r,jD);
q       = QLVLH(r,v);
bBody   = QForm(q,bECI);
uSun    = SunV1(jD);

TimeHistory(t,bBody*1e9,{'B_x (nT)' 'B_y (nT)' 'B_z (nT)'}, 'B Body');

n = length(t);
torqName = {'Residual Dipole' 'Solar'};
torq = zeros(3,n,3);

for k = 1:n
  torq(:,k,1) = Cross(resDipole,bBody(:,k));
  torq(:,k,2) = SolarTorque(uSun(:,k),q(:,k),dSol);
end

%% Output
OutputDisturbances;

%% Solar torque
function torque = SolarTorque(uSun,q,d)
p = 1367/3e8;
tW = zeros(3,1);
for k = 1:4
  angle = acos(Unit(uSun([1 3]))'*Unit(d.uWing([1 3],k)));
  s     = sin(angle);
  c     = cos(angle);
  u     = [c 0 -s;0 1 0;s 0 c]*d.uWing(:,k);
  f     = SolarF( p, d.sCell, u, uSun, d.areaSP );
  fB    = QForm(q,f);
  tW    = tW + cross(d.rWing(:,k)-d.c,fB);
end

tB = zeros(3,1);
for k = 1:6
  uSB    = QForm(q,uSun);
  if( Dot(d.uBus(:,k), uSB) > 0 )
    f      = SolarF( p, d.sBus(:,k), d.uBus(:,k), uSB, d.areaBus );
    f      = QForm(q,f);
    tB     = tB + cross(d.rBus(:,k)-d.c,f);
  end
end

torque = tB + tW;

end
```



Example 21.1: Disturbances on a communications satellite during summer solstice. The Sun is 23.5 degrees out of the orbit plane.

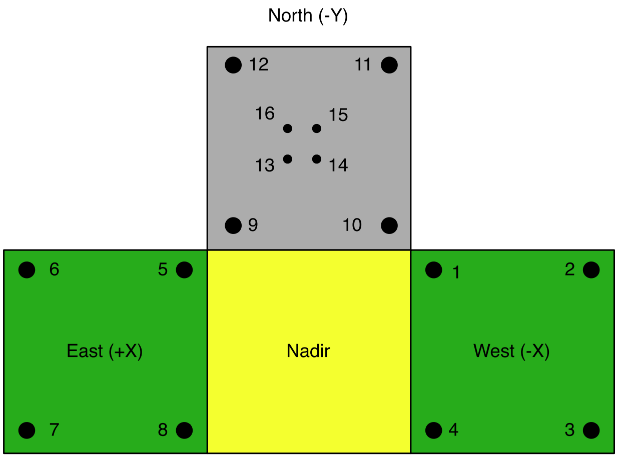


Figure 21.4 Thruster layout diagram.

21.12. Acquisition using the dual-spin turn

21.12.1 Dynamics

The dual-spin turn is based on the conservation of angular momentum. Inertial angular momentum is conserved, therefore the momentum vector will remain fixed in the inertial frame unless an external torque is applied. The spacecraft is first precessed so that the $+z$-axis (the spin-axis) is aligned with positive orbit normal. The momentum wheel that is normal to the $+z$-axis is turned on. As it spins up momentum is transferred in the body frame to the $-y$-axis. Since the angular-momentum vector is inertially fixed, the $-y$-axis aligns itself with a positive orbit normal with some residual spin rate about the $-y$-axis. For the dual-spin turn to work, the initial spin rate about the $+z$-axis must be within the limits specified by the equations

$$\frac{hW}{I_{Yaw}} \leq \Omega \leq h_W \frac{1-f}{fI_{Yaw}} \tag{21.3}$$

$$f = \left| 1 - \frac{I_{Pitch}}{\max(I_{Yaw}, I_{Roll})} \right| \tag{21.4}$$

where I_{Pitch} is the inertia of the axis with which the wheel is aligned, I_{Yaw} is the inertia of the axis about which the spacecraft is initially spinning, and I_{Roll} is the other axis inertia [1].

21.12.2 Actuators and sensors

The momentum-wheel motor is the only actuator used during the dual-spin turn. No sensing is required during the dual-spin turn. Fuel slosh provides sufficient nutation damping to eliminate the need for a nutation damper or thruster firings to damp nutation.

21.12.3 Initialization

The dual-spin turn is initialized by despinning the spacecraft to the desired spin rate and then commanding the momentum wheel to the desired postdual-spin turn rate. The spin rate is chosen so that the total momentum of the spinning spacecraft can be absorbed by the momentum wheel.

21.12.4 Simulation

A simulated dual-spin turn is shown by typing DST([0;0;.1], 500, 'Comsat', 2). The rate about the z-axis goes to zero and the momentum is absorbed in the momentum wheel and the y-axis nutation damps to zero by the end of the simulation. In the second plot, the upper line is the commanded momentum. Fuel slosh is modeled using a damper wheel in this simulation, see Fig. 21.5.

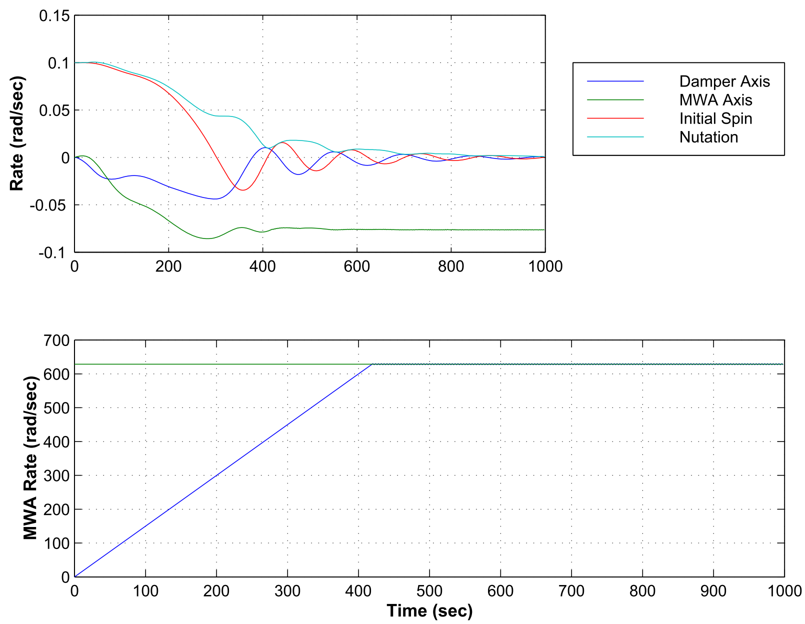


Figure 21.5 Dual-spin turn.

21.12.5 Pitch acquisition

Once the spacecraft is spinning about its y-axis and nutation has damped, the pitch-control system is activated and thrusters are selected. The Earth-sensor processing will automatically estimate pitch rate and pitch even though the Earth is only passing through the sensor field-of-view.

21.13. Dynamics

21.13.1 Introduction

During the mission orbit, a momentum-bias spacecraft is modeled as a rigid core with a rotor and flexible solar arrays. During normal operation, the solar-array flexibility does not impact the control-system performance. Consequently, it is convenient to discuss the dynamics of the two operational modes separately.

21.13.2 Normal operations

The roll/yaw and pitch dynamics can be decoupled for analysis of normal operations. The pitch dynamics are a double integrator. The roll dynamics are a coupled fourth-order plant. There are two purely imaginary pole pairs. The first at orbit rate is due to the kinematics, the second is the nutation mode, and is due to the bias momentum. The main diagonal channels each have a purely imaginary zero pair. This zero pair is located between the orbit rate and nutation mode providing 180 degrees of phase shift and making the control problem easier.

Disturbances at orbit rate, or the nutation frequency, will cause uncontrolled growth in the attitude errors. Since many of the disturbance sources are at orbit rate, it is necessary to add an automatic control system. The nutation pole causes large oscillatory responses whenever a sudden torque is applied to the spacecraft.

The high- and low-frequency limits are of interest. At high frequencies as s approaches infinity, the system becomes a double integration. At low frequencies, the orbit-rate pole pair dominates. The attitude kinematics come from the small-angle approximation

$$\omega = \dot{\theta} + (1 - \theta^{\times})\nu \tag{21.5}$$

where ν is the orbit rate vector. The dynamical equations are

$$I\dot{\omega} + \omega^{\times}(I\omega + h_W) + \dot{h}_W = T \tag{21.6}$$

For the purpose of analyzing the roll/yaw dynamics assume that h_W is constant. The orbit-rate vector is $\begin{bmatrix} 0 & -\omega_o & 0 \end{bmatrix}^T$. Assume also that the inertia matrix is diagonal. Linearizing the dynamics, and coupling with the kinematics gives the transfer-function matrix

$$\begin{bmatrix} \theta_x(s) \\ \theta_y(s) \end{bmatrix} = \frac{\begin{bmatrix} I_z s^2 + \omega_o h_z & (\omega_o I_x - h_x)s \\ -(\omega_o I_z - h_z)s & I_x s^2 + \omega_o h_x \end{bmatrix}}{I_x I_z (s^2 + \omega_o^2)\left(s^2 + \left(\frac{h_x h_z}{I_x I_z}\right)\right)} \begin{bmatrix} T_x/I_x \\ T_x/I_x \end{bmatrix} \tag{21.7}$$

$$h_x = h_w + \omega_o(I_y - I_z)$$
$$h_z = h_w + \omega_o(I_y - I_x)$$

The roll/yaw equations and the equation of the damper wheel are

$$\begin{aligned} I_x\dot{\omega}_x + \omega_y\omega_z(I_z - I_y) - \omega_z h_w + J(\dot{\Omega} + \dot{\omega}_x) &= T_x \\ I_z\dot{\omega}_z + \omega_y\omega_x(I_y - I_x) - \omega_y(\Omega + \omega) + \omega_x h_w &= T_z \\ J(\dot{\Omega} + \dot{\omega}_x) &= -D\Omega \end{aligned} \tag{21.8}$$

where J is the inertia of the damper wheel and D is the damping coefficient. If we lump J with I_x we can simplify the above to

$$\begin{aligned} \dot{\omega}_x + k_x\omega_z + \epsilon\dot{\Omega} &= 0 \\ \dot{\omega}_z + k_z\omega_x - \gamma\dot{\Omega} &= 0 \\ \dot{\Omega} + \dot{\omega}_x &= \sigma\Omega \end{aligned} \tag{21.9}$$

where

$$k_x = \frac{\omega_y(I_z - I_y) - h_w}{I_x} \tag{21.10}$$

$$k_z = \frac{\omega_y(I_y - I_x) + h_w}{I_z} \tag{21.11}$$

$$\gamma = \frac{\omega_y J}{I_z} \tag{21.12}$$

$$\sigma = \frac{D}{J} \tag{21.13}$$

$$\epsilon = \frac{J}{I_x} \tag{21.14}$$

The characteristic equation can be written in Evans form

$$\frac{s^2 - \left(\frac{k_x(k_z+\gamma)}{1-\epsilon}\right)}{s^2 - k_x k_z} = -\frac{\sigma}{1-\epsilon} \tag{21.15}$$

When $\sigma = 0$ the numerator polynomial gives the poles of the system. When $\sigma = \infty$ the denominator gives the poles of the system. Hence, the open-loop zeros are given by the denominator and the open-loop poles by the numerator. The $s = 0$ pole is the damper-wheel pole and the quadratic terms give the roll/yaw poles. For stability, the open-loop zeros must be between the closed-loop poles. These equations are only valid for $J \neq 0$.

$$\begin{bmatrix} \theta_x(s) \\ \theta_y(s) \end{bmatrix} = \frac{\begin{bmatrix} 1/I_x & 0 \\ 0 & 1/I_z \end{bmatrix}}{s^2} \begin{bmatrix} T_x \\ T_z \end{bmatrix} \tag{21.16}$$

As is evident from the transfer-function matrix, high-bandwidth controllers can ignore the orbit rate and nutational dynamics. At low frequencies the plant becomes

$$\begin{bmatrix} \theta_x(s) \\ \theta_y(s) \end{bmatrix} = \frac{\begin{bmatrix} \omega_o & (\omega_o I_x/h_x - 1)s \\ (1 - \omega_o I_z/h_z)s & \omega_o \end{bmatrix}}{s^2 + \omega_o^2} \begin{bmatrix} T_x/h_x \\ T_z/h_z \end{bmatrix} \tag{21.17}$$

The diagonal terms are small and the dynamics are dominated by crosscoupling torques. Controllers concerned with attenuating low frequency (near orbit rate) can ignore the nutation mode.

21.13.3 Dual-spin stability

A dual-spin spacecraft should be passively stable as long as the momentum wheel is at its nominal speed. This can be demonstrated by assuming that energy dissipation is provided by a damper wheel.

21.13.4 Station-keeping operations

It is necessary to include the dynamics of flexible appendages, mainly the solar arrays, for station-keeping operations. There are two reasons for this. The first is that station-keeping control systems have to respond to large disturbances caused by the delta-V thrusters. Since it is desirable to minimize the average pointing error during the burn the station-keeping system should respond quickly to these disturbances. This leads to a high-bandwidth system that may interact with flexible modes of the spacecraft. The second reason is that all thrusters are nonlinear. Even throttleable thrusters have limited throttle range and a significant minimum impulse bit. The nonlinearities give rise to control spillover beyond the control-system bandwidth. Consequently, it is necessary to consider higher-frequency modes.

The simplest way to add flex modes is to assume that the spacecraft's center-of-mass does not move when the spacecraft bends. In addition, assume that the flexible part of the spacecraft is not rotating relative to the core of the spacecraft. Finally, assume that the displacements due to bending are small and that nonlinear terms, which include the bending displacement and rate, are insignificant.

The geometry of the point mass with respect to the center-of-mass is illustrated in Fig. 21.6.

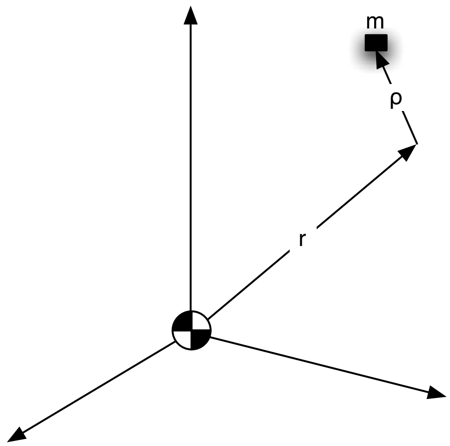


Figure 21.6 Station-keeping geometry.

Without any loss of generality, the flexible-body model consists of point masses. r is the vector from the center-of-mass to the point mass is not necessarily small. ρ is the displacement of the mass from its nominal position.

We will assume that the inertia matrix of the spacecraft includes the contributions due to the mass m and the moment arm r. Thus I is the inertia of the undeformed spacecraft. The equation for the core spacecraft angular acceleration is

$$I\dot{\omega} + \omega^{\times}(I\omega + h_w) + \dot{h}_w + \sum m_i r_i^{\times}\ddot{\rho}_i = T \tag{21.18}$$

The summation is over the physical coordinates. The acceleration of each point mass causes a torque on the spacecraft. The equation for the point-mass acceleration is

$$\ddot{\rho}_i = r_i^{\times}\dot{\omega} + \omega^{\times}\omega^{\times} r_i = \frac{f_i}{m_i} - \frac{F}{m_i} \tag{21.19}$$

These equations are quite general and can apply to any flexible appendage of any shape or size. Since the center-of-mass does not move, there is no inertial coupling between the flexible degrees-of-freedom. Coupling may be introduced through the force term on the right-hand side.

For simple systems these equations can be solved directly. For more complex systems with many flexible degrees-of-freedom it is convenient to transform the flexible degrees-of-freedom into modal coordinates. Neglect the nonlinear terms and write

$$\ddot{\rho}_i = r_i^{\times}\dot{\omega} = -k_i\rho_i - F \tag{21.20}$$

Substitute $\rho = \Phi\eta$, where Φ is the modal transformation matrix and premultiply by Φ^T

$$\Phi^T m_i \Phi \eta_i - m_i r_i^{\times} \dot{\omega} = \Phi^T k_i \Phi \eta_i - F \tag{21.21}$$

Choose Φ so that

$$\Phi^T m_i \Phi = E \tag{21.22}$$

$$\Phi^T k_i \Phi = \mathrm{diag}(\sigma_i^2) \tag{21.23}$$

This decouples the flexible degrees-of-freedom. It also makes it easier to reduce the order of the model because it can be done based on both the frequency of the mode and its modal participation.

Example 21.2 shows a station-keeping control system with flexible modes. This only shows the roll axis.

The eigenvalues are shown in the output in the command window. Only the first four are shown.

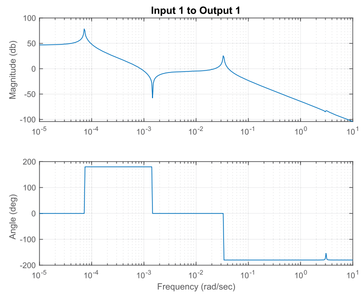
```
Open loop eigenvalues with the array at 0 deg
   0.0000e+00 + 0.0000e+00i
   0.0000e+00 + 0.0000e+00i
   0.0000e+00 + 7.2921e-05i
   0.0000e+00 - 7.2921e-05i

Closed loop eigenvalues
   0.0000e+00 + 0.0000e+00i
   0.0000e+00 + 0.0000e+00i
  -2.9060e+00 + 0.0000e+00i
  -2.4347e+00 + 0.0000e+00i
```

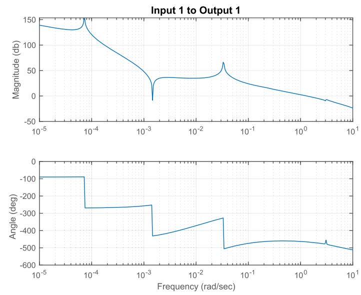
21.13.5 Actuators and sensors

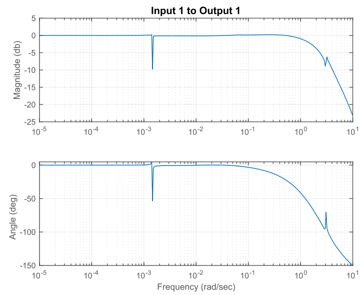
There are two sensors used during the mission orbit. One is the Earth sensor that measures roll and pitch. The second is the rate-integrating gyro that gives integrated rates as the output. No measurement of absolute yaw attitude is available. The Earth sensor is used to measure roll and pitch during all mission-orbit modes. During station keeping, the yaw gyro is used to measure yaw. This permits a higher-bandwidth roll loop for more precise control.

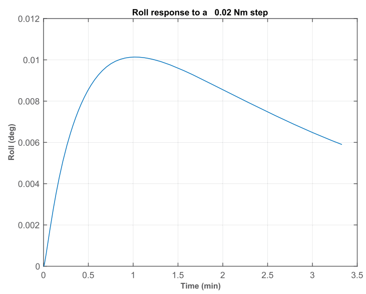
Thrusters, magnetic torquers, and the momentum-wheel motor are used for control. Alternatively, a single- or double-pivoted momentum wheel could be used and the pivot motors used for control. Magnetic torquers depend on the Earth's magnetic-field vector for their control torques. The magnetic field at geosynchronous altitudes is influenced

```
%% Compensate the flexible solar array

%% Constants
degToRad = pi/180;
radToDeg = 180/pi;
dT       = 0.25;
nSim     = 800;
uStep    = 0.02;
rBModel  = false;


%% Database
inr = ComStar('MO Inertia');

%% The state is [qX;qY;qZ;wX;wY;wZ;modes]
w = logspace(-5,1,400);

%% Load the flex model
load FlexM00

%% Open loop eigenvalues
disp('Open loop eigenvalues with the array at 0 deg');
disp(eig(a));

d     = 0;
n     = size(b,1);
b     = b(:,1); % Tx
c     = [1 zeros(1,n-1)];

FResp(a,b,c,d,1,1,w,'unwrap');
Rename('Roll Open Loop');

%% Design the PID
zeta    = 4;
omega   = 0.3;
tauInt  = 200;
omegaR  = 10*omega;

[aC, bC, cC, dC] = PIDMIMO( inr(1,1), zeta, omega, tauInt, omegaR );

if( rBModel )
  a = [0 1;0 0];
  b = [0; 1/inr(1,1)];
  c = [1 0];
  d = 0;
end

%% Append the plant and the controller
[a,b,c,d]        = Series(a,b,c,d,aC,bC,cC,dC);

FResp(a,b,c,d,1,1,w,'unwrap');
Rename('Roll Open Loop with Compensator');

aCL = a - b*c;
FResp(aCL,b,c,d,1,1,w,'unwrap');
Rename('Roll Closed Loop with Compensator');

%% Display the eigenvalues
s = eig(a - b*c);
DispWithTitle(s,'Closed loop eigenvalues');

%% Simulate the loop
yPlot    = zeros(1,nSim);
n        = length(aCL);
x        = zeros(n,1); % Rest conditions at the start
c        = [1 zeros(1,n-1)];

[aCL, b] = C2DZOH( aCL, b, dT );

for k = 1:nSim
    yPlot(k) = c*x;
    x        = aCL*x + b*uStep;
end

titleStr = sprintf('Roll response to a %6.2f Nm step',uStep);

t = (0:(nSim-1))*dT;
TimeHistory(t,yPlot*radToDeg,{'Roll (deg)'},titleStr);

Figui;
```









Example 21.2: Roll response for a station-keeping control system.

primarily by the Sun. As a consequence, it varies diurnally and with the solar cycle. During periods of intense solar activity, the field can even reverse direction.

The main advantage of magnetic torquers is that they do not change the satellite's velocity and they do not consume any propellant. Of course, their mass, power, telemetry, and command impacts must be considered. It is often better to use thrusters for momentum unloading and roll/yaw control.

The momentum-wheel motor is used to control the pitch axis. If there are constant-pitch torques, the momentum wheel will eventually saturate and the momentum-wheel motor will be unusable until momentum is unloaded with thrusters.

Thrusters are used to change the velocity of the spacecraft and are used when either the magnetic torquers or momentum-wheel motor cannot be used. This can happen during magnetic-field disturbances or when the errors are too large for the torquers or motor to control. The disadvantage of using thrusters is that they consume fuel and that the control threshold for chemical thrusters (e.g., hydrazine) is quite large, making precise control difficult.

21.13.6 Control-system organization

The flow of the control system is

1. Sensor-interface processing;
2. Error computation;
3. Control-torque computation;
4. Control-torque distribution;
5. Actuator-interface processing.

The interface processing is the low-level conversion to and from hardware data. The error computation takes the raw sensor data (now in physically meaningful form) and subtracts biases, misalignments, etc. to produce control system errors. These are processed by the control laws to produce desired torques. The control torques are converted to pulsewidths, in the case of thrusters and magnetic torquers, and voltage, in the case of the momentum-wheel motor, by the control-torque distribution algorithms. The actuator-interface processing then converts the results into values that the hardware can read.

21.13.7 Modes

The mission-orbit control system has two modes

1. Low-bandwidth control;
2. High-bandwidth control.

Either may be implemented with the momentum-wheel motor and magnetic torquers or with thrusters. The former mode uses only the Earth sensor for sensing, while the latter mode requires the yaw gyro. When thrusters are used, the momentum wheel automatically goes to a preset speed (which can be its current speed). The high-bandwidth control is normally used for station keeping. The low-bandwidth mode is used with thrusters to transition from station keeping to torquer/momentum-wheel motor control, or when the magnetic field is perturbed.

21.13.8 Earth sensor

Earth sensors are of three types, conical scanning, scanning, and static. The latter two are the most popular on geosynchronous satellites. A major advantage of scanning sensors is that they give roll and pitch outputs directly. The static Earth sensor has sets of thermopiles arranged in a circle about the boresight. Typically each set has three. One looks at the Earth all of the time, the second straddles the Earth, and the third looks at cold space. The former and latter are used to calibrate the straddling sensor. The calibrated temperatures of the straddling sensors are used to compute roll and pitch. The scanning sensor is quite a bit simpler to incorporate into a system so it was selected for this design.

21.13.9 Gyros

A mechanical rate-integrating gyro was chosen for the gyro. This kind of gyro is inexpensive, has the required life (if only used for station keeping), and is reasonably accurate. Low drift rates are not required for this application since the gyros are only used for station-keeping maneuvers that rarely last more than 30 minutes. Each gyro has two outputs. When it senses a zero rate it generates a 44-kHz pulse train on each line. The two lines are attached to a counter, one to the upcount and one to the downcount. When a 1 least significant bit (LSB) positive rate is sensed the upcount line pulse count increases by one pulse per cycle and the downcount line pulse count decreases by one pulse per cycle. Hence, the threshold for rate detection is equal to two pulses. The up–down counter is read at 4 Hz and the difference between the current and previous values of the counter gives the change in attitude over the sampling interval. Counter overflow must be accounted for in the computation. The integrated change in angle is then passed through a noise filter and used for yaw rate and attitude. The gyro must be initialized by passing it through a low-pass filter during the gyro-initialization phase.

21.13.10 Noise filtering

All sensors have significant amounts of random noise that should be filtered before use by the control system. This filtering is independent of any filtering that is done by the control loops themselves.

The sensors are sampled at 4 Hz. To get the maximum filtering, the noise filters are designed for a 4-Hz sampling frequency. To minimize the impact on the control system no more than 10° of phase shift is acceptable at 0.1 rad/s.

21.13.11 Momentum-wheel pitch and tachometer loops

The basis for the pitch control is a tachometer loop that maintains the speed of the momentum wheel. It is desirable to keep the wheel speed within a range of the nominal speed, normally ±10% to keep the spacecraft gyroscopically stiff. A DC motor has the transfer function

$$\frac{\omega}{T} = \frac{1}{Js + \beta} \tag{21.24}$$

where β is due to the back-emf and viscous friction in the motor. A simple control scheme is to multiply the difference between the desired speed and the measured speed by a gain K and to filter the measured speed by a first-order filter. The resulting closed-loop system is

$$\Omega = \frac{(s + \omega_T)(K\Omega_C + T|C)}{Js^2 + (J\omega_T + \beta)s + \omega_T(K + \beta)} \tag{21.25}$$

where ω_T is the filter cutoff, ω_T is chosen to provide adequate damping, and K is made sufficiently large to provide good disturbance rejection and command tracking.

The input to the tachometer loop is the wheel-speed demand. The equation for the pitch loop is

$$I\ddot{\theta} + J\dot{\omega} = T \tag{21.26}$$

Since we want a pitch equation of the form

$$I\ddot{\theta} + c\dot{\theta} + k\theta = T \tag{21.27}$$

it follows that the wheel speed demand will be

$$\Omega = \frac{c\theta + k\int\theta}{J} \tag{21.28}$$

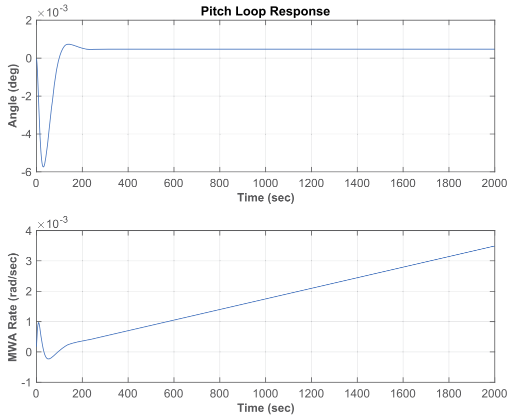
which is a proportional-integral (PI) controller.

Example 21.3 shows the pitch and tachometer loop response. The step response causes the wheel to increase in speed. Eventually, it will need to be unloaded with thrusters.

21.13.12 Low-bandwidth roll/yaw control

There are two aspects of the roll/yaw control system that must be considered. The first is that it must attenuate the external disturbances on the spacecraft. These are at harmonics of the orbit rate and generally have no significant components above twice the orbit rate. The second aspect is nutation damping. Thruster firings will tend to excite nutation. The control system must damp the nutation. Since the nutation

```
%% Tests the MWA pitch loop
inr          = ComStar('MO Inertia');
inrMWA       = ComStar('MWA Inertia');

%% Design the pitch and tach loops
zeta         = 0.7071;
wN           = 0.2;
tSamp        = 0.25;
zetaPL       = 0.7071;
wNPL         = 0.04;
beta         = 0.0;
PitchLoop(inr(2,2),inrMWA,beta,zeta,zetaPL,wN,
          wNPL,1,'Z');
```



Example 21.3: Pitch-loop response.

frequency is much higher than the orbit rate we can break the design process into two parts, one to attenuate low-frequency disturbances and the second to damp nutation.

The controller will be designed in two parts. The first part takes the low-frequency approximation to the open-loop system and selects a pair of gains to meet the pointing requirements. This approach, a purely proportional control, is the simplest. An alternative is to estimate yaw and use full-state feedback. This system is one input and two outputs but the two outputs are always in a fixed ratio to each other.

The second part is the nutation damper. We will insert a second-order compensator to stabilize nutation.

The first step is to reduce the roll/yaw equations to their low-frequency approximation. The roll/yaw equations are

$$-\frac{T_z}{h_w} = \dot{\theta}_x - \omega_o\theta_z \frac{T_x}{h_w} = \dot{\theta}_z + \omega_o\theta_x \tag{21.29}$$

Remember that

$$h = \begin{bmatrix} 0 \\ -h_w \\ 0 \end{bmatrix} \quad \nu = \begin{bmatrix} 0 \\ -\omega_o \\ 0 \end{bmatrix} \tag{21.30}$$

The torque command is

$$\begin{bmatrix} T_x \\ T_z \end{bmatrix} = h_w \begin{bmatrix} K_{xx} \\ K_{zx} \end{bmatrix} \theta_x \tag{21.31}$$

since only roll is measured. This can be implemented with either a skew dipole or separate torquers since the ratio of the torques is fixed. Alternatively, we could have used a dynamic compensator by having

$$\begin{bmatrix} T_x \\ T_z \end{bmatrix} = h_w \begin{bmatrix} K_{xx} & K_{xz} \\ K_{zx} & K_{zz} \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} \tag{21.32}$$

and estimating yaw. The latter approach requires two separate magnetic torquers.

The transfer function for the first approach is quite simple and is

$$\begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \frac{1}{h_w} \frac{\begin{bmatrix} s & \omega_o \\ -\omega_o + K_{xx} & s + K_{zz} \end{bmatrix}}{s^2 + K_{zx}s + \omega_o^2 - \omega K_{xx}} \begin{bmatrix} -T_z \\ T_x \end{bmatrix} \tag{21.33}$$

The disturbance torques are scaled by the bias momentum, therefore large bias momentum will reduce the attitude errors. K_{zx} provides damping. K_{xx} must be negative if it is larger in magnitude than ω_o, which will almost always be the case. The major disturbance sources will be at zero frequency and orbit rate. The transfer-function matrices are

$$\begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \frac{1}{h_w} \frac{\begin{bmatrix} 0 & \omega_o \\ -\omega_o + K_{xx} & K_{zz} \end{bmatrix}}{\omega_o^2 - \omega K_{xx}} \begin{bmatrix} -T_z \\ T_x \end{bmatrix} \tag{21.34}$$

The magnitude of the errors for large gains is

$$\begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \frac{1}{h_w} \frac{T}{\omega_o h_w \sqrt{K_{xx}^2 + K_{zx}^2}} \begin{bmatrix} 2\omega_o \\ |K_{xx}| + K_{zx} \end{bmatrix} \tag{21.35}$$

increasing either gain will decrease the roll errors. However, the two gains must be carefully balanced to attenuate yaw errors.

The estimator approach is to use full-state feedback and to estimate yaw. The estimator is

$$\begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_z \end{bmatrix} = \begin{bmatrix} 0 & \omega_o \\ \omega_o & 0 \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} + \begin{bmatrix} L_x \\ L_z \end{bmatrix} (z - \theta_x) \tag{21.36}$$

The transfer functions are

$$\frac{\theta_x}{z} = \frac{sL_x + \omega_o L_x}{s^2 + sL_x + \omega_o(\omega_o + L_z)} \tag{21.37}$$

$$\frac{\theta_z}{z} = \frac{sL_z - \omega_o L_x}{s^2 + sL_x + \omega_o(\omega_o + L_z)} \tag{21.38}$$

and at $s = j\omega_o$ these become

$$\frac{\theta_x}{z} = 1 \tag{21.39}$$

$$\frac{\theta_z}{z} = j \tag{21.40}$$

At orbit rate, the estimator always passes the roll measurement unfiltered to the roll estimate and phase shifts the roll measurement by 90 deg to get the yaw estimate. Note that the estimator rolls off as $1/s$, making it sensitive to unmodeled dynamics. The roll to yaw transfer function is a nonminimum phase. This limits the gain of any controller using this estimator.

Example 21.4 shows the low-frequency roll/yaw control. The last two plots show the controller performance using the disturbance model generated in this chapter.

The next step is to compensate for the nutation. First, turn the one-input–two-output system into a SISO system by appending the gain calculated above.

Several different approaches can be used to stabilize nutation. One is to use a nonminimum phase-transfer function of the form

$$\frac{s-a}{s+a} \tag{21.41}$$

The magnitude of this transfer function is always one and the phase is

$$\phi = \tan^{-1}\left(\frac{-2\omega}{\omega^2 - a^2}\right) \tag{21.42}$$

at large ω the phase is $\tan^{-1}(-0)$ or -180°. Since the magnitude of the phase shift is greater than 90°, this is known as a nonminimum phase-transfer function. Interestingly, the first-order Padé approximant to a delay of period T is

$$\frac{s-2/T}{s+2/T} \tag{21.43}$$

Thus this nonminimum phase-transfer function could be implemented as a delay.

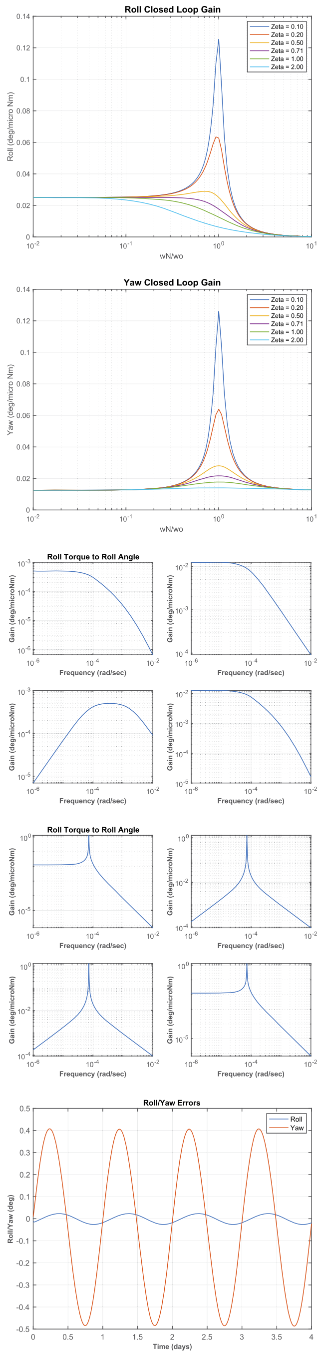
The approach presented here is to introduce a complex zero just before the nutation pole pair using the transfer function

$$\frac{s^2+\omega_z^2}{s^2+2\zeta\omega_p+\omega_p^2} \tag{21.44}$$

and a damped pole after the nutation mode. As long as the zero is at a frequency less than that of the nutation pole pair the system will be stable. Example 21.5 shows nutation damping.

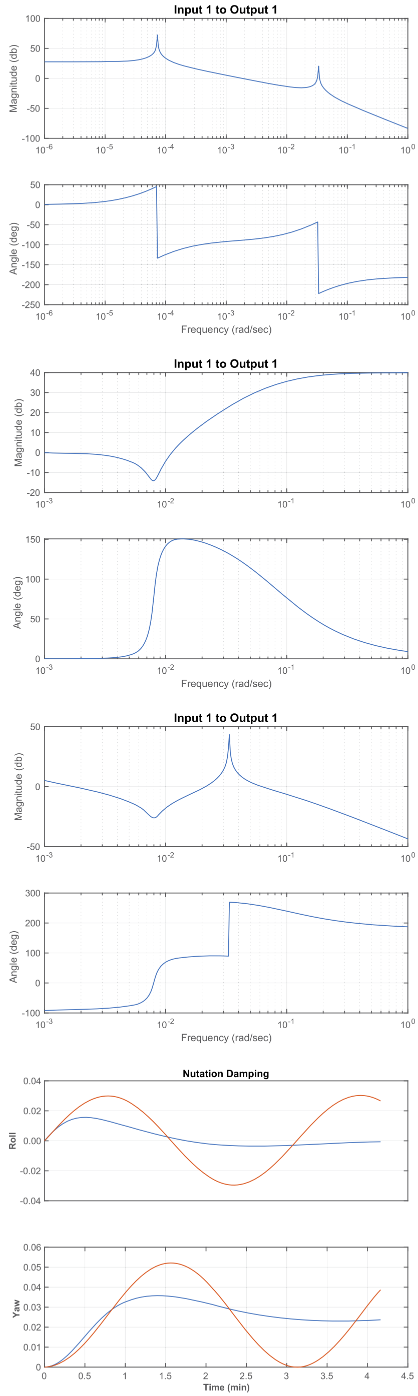
```
Eigenvalues
   -6.4909e-02  +  7.2632e-02i
   -6.4909e-02  -  7.2632e-02i
```

```
degToRad      = pi/180;
radToDeg      = 180/pi;

wo            = 2*pi/(24*3600);
hW            = ComStar('MWA Inertia')*ComStar('Nominal MWA Rate');
iAxis         = [1 3];
inr           = ComStar('MO Inertia');
hMWA          = ComStar('U MWA')*hW;
wO            = ComStar('Orbit Rate Unit Vector')*wo;
bMin          = 75e-9;

%% Plant models
[aP,bP,cP,dP,aL,bL,cL,dL] = RYDyn( inr, hMWA, wO, iAxis);

n    = 100;
nZ   = 6;
zeta = [0.1 0.2 0.5 0.7071 1.0 2.0];
k    = logspace(-2,1,n);
j    = sqrt(-1);
wN   = k*wo;
kX   = (wo^2-wN.^2)/wo;
f0   = 1.e-6/(wo*hW);
kZ   = zeros(nZ,n);
aX   = zeros(nZ,n);
aZ   = zeros(nZ,n);
leg  = cell(nZ,1);
for i = 1:nZ
  kZ(i,:) = 2*zeta(i)*wN;
  f       = f0./(j*kZ(i,:) - kX);
  aX(i,:) = abs( 2*wo.*f );
  aZ(i,:) = abs((j*wo + kZ(i,:) - j*(kX-wo)).*f);
  leg{i}  = sprintf('Zeta = %4.2f',zeta(i));
end


Plot2D(k,aX*180/pi,'wN/wo','Roll (deg/micro Nm)','Roll Closed Loop Gain','xlog')
legend(leg);
Plot2D(k,aZ*180/pi,'wN/wo','Yaw (deg/micro Nm)','Yaw Closed Loop Gain','xlog')
legend(leg);
Plot2D(k,kZ,'wN/wo','kZ','Roll Angle to Yaw Torque Gain','xlog')
legend(leg);
Plot2D(k,kX,'wN/wo','kX','Roll Angle to Roll Torque Gain','xlog')

%% Skew dipole control system simulation
zeta  = 2.5;
wN    = 5*wo;
kRY   = [(wo^2-wN^2)/wo;2*zeta*wN];
aCL   = aL + bL*kRY*[1 0]*hW;
s     = eig(aCL);

dModel =  load('DisturbanceModel');

[a,b] = C2DZOH(aCL,bL,dModel.dT);

%% The torque transmission plots
titlesRY = {'Roll Torque to Roll Angle'...
            'Roll Torque to Yaw Angle' ...
            'Yaw Torque to Roll Angle' ...
            'Yaw Torque to Yaw Angle'};

wP          = logspace(-6,-2,300);
mag         = MagPlot(aCL,bL,cL,dL,1:2,1:2,wP);
TTPlots(radToDeg*1.e-6*mag,wP,titlesRY,'deg/microNm','Closed Loop')
[mag,io] = MagPlot(aL,bL,cL,dL,1:2,1:2,wP);
TTPlots(radToDeg*1.e-6*mag,wP,titlesRY,'deg/microNm','Open Loop')

x       = [0;0];
nSim    = size(dModel.totalTorque,2);
xPlot   = zeros(length(x),nSim);
uPlot   = zeros(2,nSim);
for k = 1:nSim
  x            = a*x + b*dModel.totalTorque([1 3],k);
  xPlot(:,k) = x;
  uPlot(:,k) = kRY*x(1)*hW/bMin;
end

rYGain = kRY*hW;

t = (0:(nSim-1))*dModel.dT;
TimeHistory(t,xPlot*180/pi,{'Roll/Yaw (deg)'},'Roll/Yaw Errors',{1:2})
legend('Roll','Yaw')
TimeHistory(t,uPlot,{'Dipole (ATM^2)'},'Torquer Demand',{1:2})
legend('Roll','Yaw')
Figui
```



Example 21.4: Low-frequency roll/yaw control.

```
%% Load in the data generated by TRYLF.m
% Which supplies a, b, c, d and rYGain.
load RYC

%% Connect the skew dipole and earth sensor
a        = aP;
b        = bP*rYGain;
c        = [1 0 0 0]; % Only roll is available
d        = 0;

w        = logspace(-6,0,400);
FResp(a,b,c,d,1,1,w);
Rename('Open Loop');

% Adjust these parameters to change the nutation damping
wz       = 0.008;
zz       = 0.1;
zp       = 1;
wp       = 0.08; %

%% The nutation compensator
[aN,bN,cN,dN]       = Gen2nd ( zz, wz, zp, wp );
w                   = logspace(-3,0,400);
FResp(aN,bN,cN,dN,1,1,w);
Rename('Nutation Compensator');

[aT,bT,cT,dT]       = Series( a, b, c, d, aN, bN, cN, dN );
FResp(aT,bT,cT,dT,1,1,w);
Rename('Open Loop with Compensator');
aCL = aT + bT*cT;
s = eig(aCL);

DispWithTitle(s,'Eigenvalues');

%% Simulate the nutation compensator
dT            = 0.25;
nSim          = 1000;
[aZCL,bZCL] = C2DZOH(aCL,bT,dT);
[aZOL,bZOL] = C2DZOH(a,b,dT);

xCL = [0;0;0.001;0;0;0]; % Some roll rate
xOL = xCL(1:4);

xCLPlot = zeros(2,nSim);
xOLPlot = zeros(2,nSim);

for k = 1:nSim
  xCLPlot(:,k) = xCL(1:2);
  xOLPlot(:,k) = xOL(1:2);
  xCL          = aZCL*xCL;
  xOL          = aZOL*xOL;
end

t = (0:(nSim-1))*dT;

TimeHistory(t,[xCLPlot;xOLPlot],{'Roll' 'Yaw '},...
  'Nutation Damping',{[1 3] [2 4]})
```



Example 21.5: Nutation damping.

```
-1.4196e-02 + 2.3776e-02i
-1.4196e-02 - 2.3776e-02i
-1.7143e-03 + 0.0000e+00i
-7.6505e-05 + 0.0000e+00i
```

21.13.13 Thruster control

Thrusters are used both during station keeping and as a backup to the magnetic-torquer/momentum-wheel control system. For backup purposes, the thruster roll/yaw control uses the same compensator as the magnetic roll/yaw control. However, pitch control uses a proportional-derivative control.

21.13.14 High-bandwidth roll/yaw and pitch control

In high-bandwidth mode, each axis is controlled independently with a proportional-integral-differential controller. The yaw measurement is taken from the yaw gyro. The yaw gyro does not provide an absolute attitude reference and the control system assumes that the initial yaw attitude is zero. This does not pose a problem as long as the mode is not run for long periods.

The rigid-body equations are

$$T = I\dot{\omega} + \omega^{\times} I\omega \tag{21.45}$$

If rates are small and the angles are small this reduces to

$$T = I\ddot{\theta} \tag{21.46}$$

If we define

$$T = Iu \tag{21.47}$$

then the axes are decoupled and the control problem is reduced to

$$u = \ddot{\theta} \tag{21.48}$$

or three decoupled double integrators. These equations are valid if the flexible modes are at much higher frequencies than the control bandwidth and, in this case, the bandwidth is much higher than the nutation mode. Once the control is computed the control torques are formed by the operation

$$Iu \tag{21.49}$$

which is a 3-by-3 matrix times a 3-by-1 vector multiply. We can now design each axis control loop independently.

A PID controller is of the form

$$T = K_p + K_R \frac{\omega_R s}{s + \omega_R} + \frac{K_I}{s} \tag{21.50}$$

The derivative term is multiplied by a first-order filter to prevent differentiation at high frequencies. Rearranging gives

$$\frac{T}{\theta} = \frac{(K_P + K_R\omega_R)s^2 + (K_p\omega_R + K_I)s + (K_P + K_i\omega_R)}{s(s + \omega_R)} \tag{21.51}$$

If ω_R is set to ∞ this becomes

$$\frac{T}{\theta} = \frac{K_R s^2 + K_P s + K_I}{s} \tag{21.52}$$

This transfer function is not proper, i.e., it goes to ∞ as s goes to 0. The latter cannot be implemented in state-space form for this reason. In practice, this is not a problem if noise filtering is included in the loop.

In this system the plant is not a set of pure double integrators, instead, it has the undamped nutation pole. If the bandwidth is set high enough, the nutation pole will be unobservable by the control system. This does not mean it has disappeared and it will be excited by control activity and the disturbance torques. This would pose a problem when the PID was turned off. The normal mode thruster control will damp the nutation and so this is not a problem.

21.13.15 Magnetic-torquer control

The magnetic torquers are controlled with a timer. For each control period, the timer is passed the duration of the magnetic pulse. The pulse always starts immediately.

21.13.16 Thruster control

The thrusters are controlled with timers. For each control period, the timer is passed the duration of each thruster pulse and a delay from the issuing of the command. This permits off-pulse modulation of the thrusters and execution of spin-precession maneuvers. The control algorithms produce a three-axis torque demand. This must be translated into pulsewidth demands for the selected thrusters. The thrusters generate unidirectional torques and it is generally not possible to divide the thrusters into bidirectional pairs. The simplest approach is to use a linear programming approach to torque distribution. This is done using the simplex algorithm. Simplex solves the problem

$$T = Au \tag{21.53}$$

$$c = \sigma_{i-1}^{n} |u_i| \tag{21.54}$$

where A is a 3-by-n matrix of torque vectors, u is an n-by-1 column vector of pulsewidth fractions, c is the cost, and T is the desired torque. Simplex finds u.

This algorithm uses matrix methods to speed the computation and is optimized for three constraint equations.

21.13.17 Actuator saturation

All actuators have a saturation limit. If the compensator has integrators this can lead to sluggish behavior. The solution is to add antiwindup compensation. This is straightforward. Implement the digital controller as

$$y_k = sat(Cx_k + Du_k) \tag{21.55}$$

$$x_{k+1} = (A - LC)x_k + (B - LD)u_k + Ly_k \tag{21.56}$$

If y is not saturated this reduces to the equations

$$y_k = Cx_k + Du_k \tag{21.57}$$

$$x_{k+1} = Ax_k + Bu_k \tag{21.58}$$

If it is saturated the dynamics of the compensator are governed by the plant $A - LD$. L must be chosen so that the saturated compensator is stable. Making $A - LD$ have all of its poles at zero is a convenient choice.

21.13.18 Thruster resolution

The pulsewidth is related to the torque demand by the equation

$$\tau = \frac{u}{u_{max}} T \tag{21.59}$$

where T is the pulsing period, u is the torque demand, and u_{max} is the maximum torque. The larger the T is the larger τ will be for a given torque demand, thus increasing the pulsing period can increase the resolution. This does not mean that the control period must be increased, rather one fires a pulse every T seconds. T should be an integral multiple of the control period.

If u is much less than u_{max} this will not normally have a major impact on the phase margins for the system. It will have an impact and must be considered when compensating the loop.

An alternative is to add triangle-wave dither to the pulse demand. If the dither frequency is high enough, and the plant has low-pass filter characteristics, dither can reduce the effect of the minimum pulsewidth. Unfortunately, in a digital control system, the sampling rate puts a limit on the dither frequency; hence the dither will appear as a high-frequency oscillation. This may not pose a problem but must be accounted for in the jitter budget. This leads to the problem of the minimum pulsewidth resolution which has a major impact on pointing performance since it leads to a large deadband around zero.

21.14. Summary

A control-system design for a momentum-bias geosynchronous communications satellite is presented. All control loops have been designed. A 13-year life and a pointing CEP of 0.17 deg meet the design specifications.

Our task was made easier by a relatively stiff array and the absence of a yaw-pointing requirement. If the array were more flexible we would have had to add compensation to the station-keeping loops to stabilize the low-frequency flex modes. If yaw were important we would have had to design a more sophisticated low-frequency yaw controller or add yaw sensing.

The next step in the design would be to add realistic actuator and sensor models and nonlinear dynamics models and verify that the design still meets requirements.

Other things to consider are:

1. Add in the actuator and sensor dynamics models.
2. Determine what commands are needed for each controller.
3. Determine how to do mode transitions (from station keeping to normal mode, for example).
4. Determine what telemetry is needed to monitor the control system.
5. Translate the MATLAB® into flight software.
6. Write operational procedures.
7. Train the spacecraft operators.

Nonetheless, this is well on the way to a complete design!

References

[1] C. Hubert, Spacecraft attitude acquisition from a spinning or tumbling state, Journal of Guidance, Control, and Dynamics 4 (2) (1981) 164–170.